

Measuring Teacher Attrition with Labour Force Survey Data

The Case of the United States

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Why this project

- Which of the factors we can observe correlate most strongly with the decision to leave teaching?
- Re-estimate the teacher leaving rate against the published figures of the official surveys.

Contribution

1. **A new series.** The longest **continuous** series of teacher attrition to reach the present: one estimate for **every year from 1997 to 2024**, constructed from twenty-eight consecutive supplements of the Current Population Survey.
2. **A measurement discussion.** A systematic treatment of the ways a longitudinal household survey identifies leaving: the linking of individuals across interviews, the alternative definitions of a leaver, and the assumptions each definition entails.
3. **Substantive evidence.** A characterization of who leaves teaching and of the observed factors most strongly associated with leaving, together with the destinations of leavers and the heterogeneity of attrition across states.

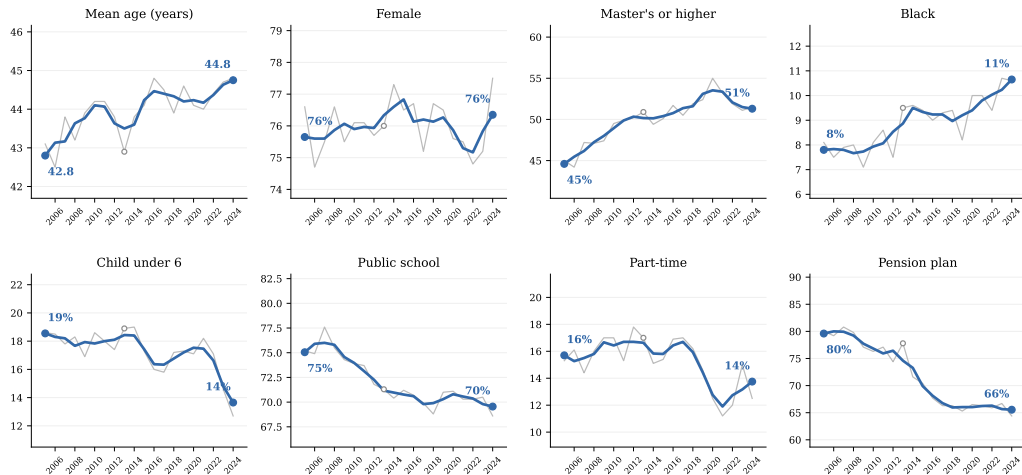
The CPS replicates the official picture — and does it every year

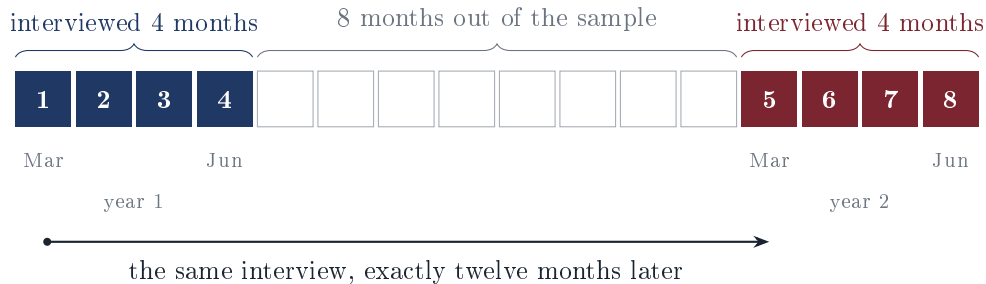
	CPS	NTPS (official)
Measurement	every year, 1997–2024	occasional waves
Example wave: 2020–21, the latest NTPS		
Female (%)	75.5	76.8
Master’s degree or higher (%)	57.0	61.0
Average age (years)	44.0	42.9
Teachers observed in the wave	3,855	39,633

Public school teachers. CPS: supplements of 2021–2022 (school years 2020–21 and 2021–22), weighted. NTPS: National Teacher and Principal Survey 2020–21 (NCES 2022-113; Digest of Education Statistics, table 209.10; respondent count from NCES 2024-024).

Full sample: **5.0 million** person-year records, 1998–2025; **104,545** teacher observations; **48,842** teachers followed across interviews in the linked panel.

And the composition is observed year by year, not wave by wave





The March supplement (ASEC)

- The largest supplement of the CPS: the March sample is expanded to roughly 90,000 households.
- Every adult reports on the *previous calendar year*: longest job held, weeks worked, earnings, benefits.
- We use it to reconstruct the leaving rate.

Two ways to measure the leaving rate

Follow the person, year to year

The rotation shows the same individual twelve months apart. The CPS has no person identifier, so records are linked on fields common to both years:

household identifier

dwelling identifier

person line number within the household

validated on sex, race,

and age advancing 0–2 years

Ask about last year, each March

The supplement's retrospective question. Whoever reports teaching as last year's longest job, and no longer teaches at the March interview, left within the year.

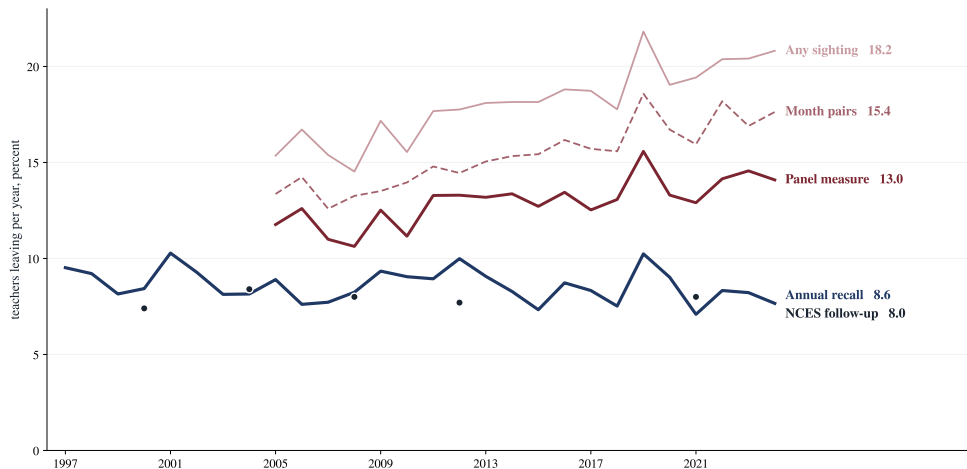
The measure used in the literature so far.

Who counts as a leaver in the linked panel?

Linking the same individuals across interview months admits many definitions. Each choice moves the rate.

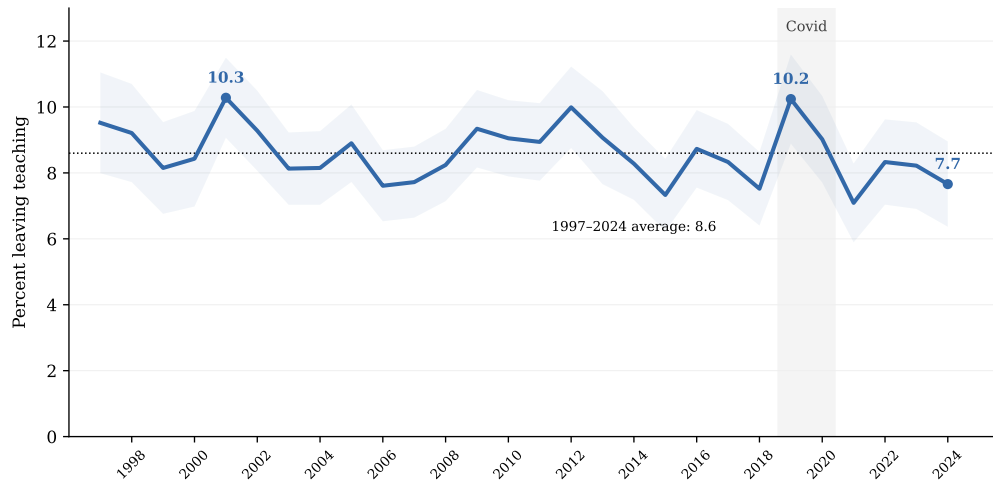
- **Panel measure (benchmark).** Teaching, with a bachelor's degree, in at least two year-1 interviews; a leaver if never observed teaching in year 2, given two or more year-2 interviews, one outside the summer.
- **Month pairs.** Every teaching month is matched to its interview twelve months ahead; one person can enter several times.
- **Any sighting.** A single teaching observation in year 1 puts the person in the denominator, including one-month spells.
- **Annual recall.** The retrospective March measure, for comparison.

One dataset, one rate per definition



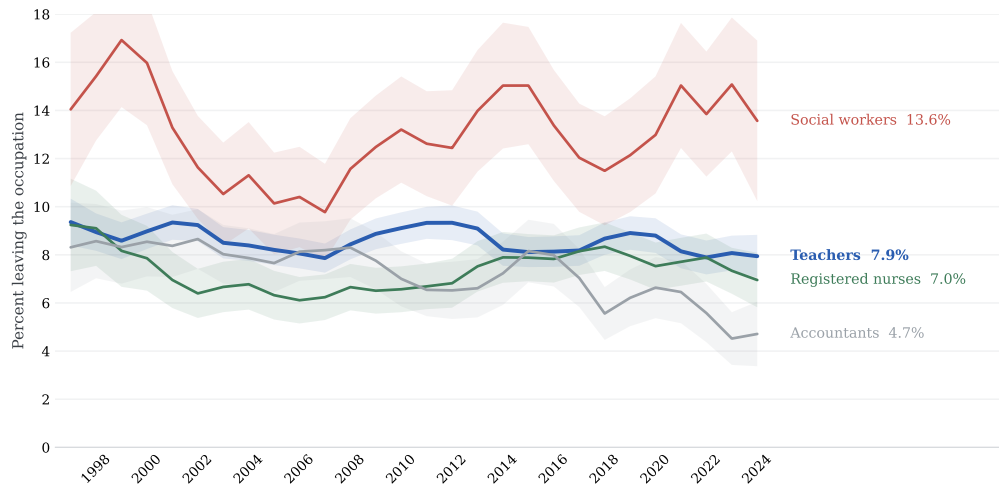
Dots: NCES Teacher Follow-up Survey, the official follow-up estimate.

The continuous series, 1997–2024



Retrospective annual measure, with 95 percent confidence bands.

The rate in context: comparable professions



What is the leaving rate for?

Not an end in itself: with person-level exits we can ask which observed factors correlate with the decision to leave.

Why teachers leave: what the CPS can and cannot see

Reasons identified in the literature	Observed in the CPS?
Working conditions	no
Organizational support (mentoring, leadership)	no
Autonomy and intrinsic motives	no
Financial factors	yes
Personal life-cycle motives	yes
Sociodemographic characteristics	yes

Five broad families of reasons from our review of the literature. The CPS covers the economic and life-cycle margins.

Five messages

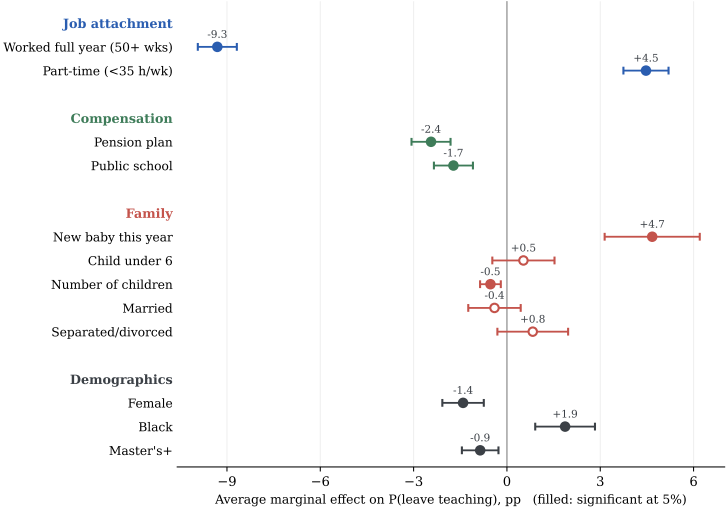
1. **Economic and employment conditions correlate with leaving far more than demographics.**
2. Weak job attachment is the strongest single correlate: part-time and part-year teachers leave at several times the rate of full-time, full-year teachers.
3. Compensation matters: pension coverage is associated with 2.4 points less attrition, and exits concentrate in the lowest earnings deciles.
4. Life-cycle events matter: childbirth raises a woman's exit probability by about 6 points, more than in comparable professions.
5. Most leavers do not move to another job: they leave employment, and the differences across states are large.

The econometric specification

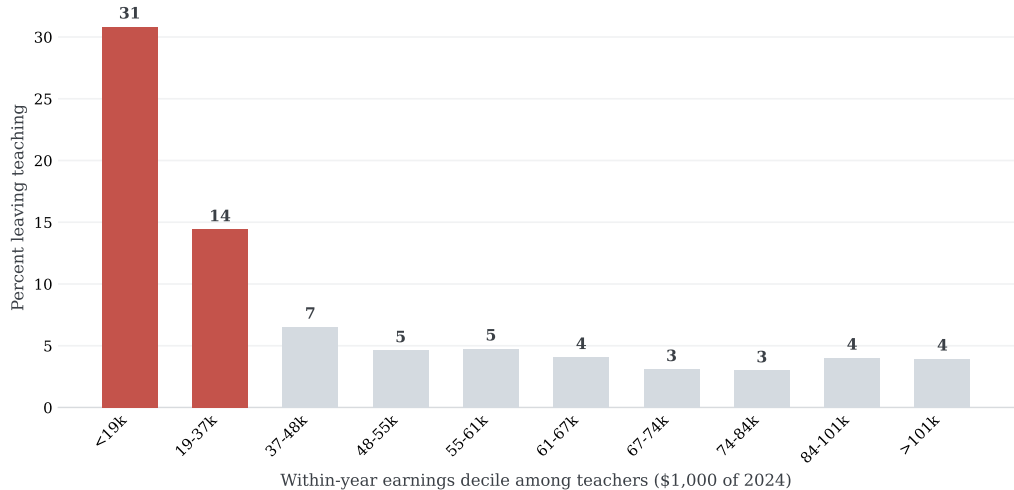
$$\Pr(\text{Leave}_{it} = 1 \mid x_{it}) = \Phi\left(\underbrace{\text{attach}'_{it}\beta_1}_{\substack{\text{full year,} \\ \text{part-time}}} + \underbrace{\text{comp}'_{it}\beta_2}_{\substack{\text{pension,} \\ \text{public school}}} + \underbrace{\text{fam}'_{it}\beta_3}_{\substack{\text{new baby, children,} \\ \text{marital status}}} + \underbrace{\text{dem}'_{it}\beta_4}_{\substack{\text{age, age}^2, \text{sex,} \\ \text{race, education}}} + \delta_t\right)$$

- Probit on college-graduate teachers, pooled 2015–2024, with calendar-year effects δ_t ; $n = 28,900$.
- Average marginal effects reported, in percentage points.
- Robust to logit and weighted LPM, state and year fixed effects, the full 1997–2024 window ($n = 85,430$), and the occupation-pair definition of leaving.

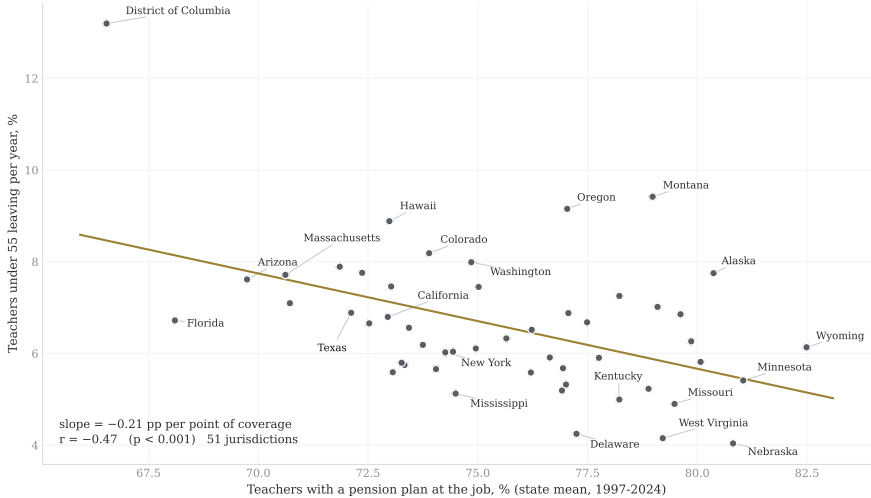
Message 1 · Economic conditions dominate demographics



Message 2 · Exits concentrate at the bottom of the pay scale

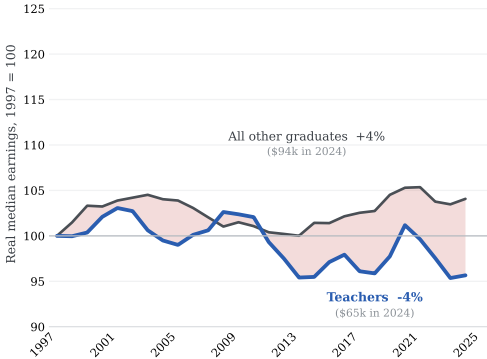


Message 2 · Where pension coverage is higher, fewer teachers leave

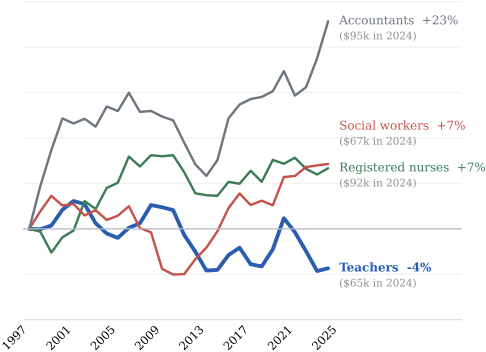


Message 2 · Teachers' relative pay has drifted down

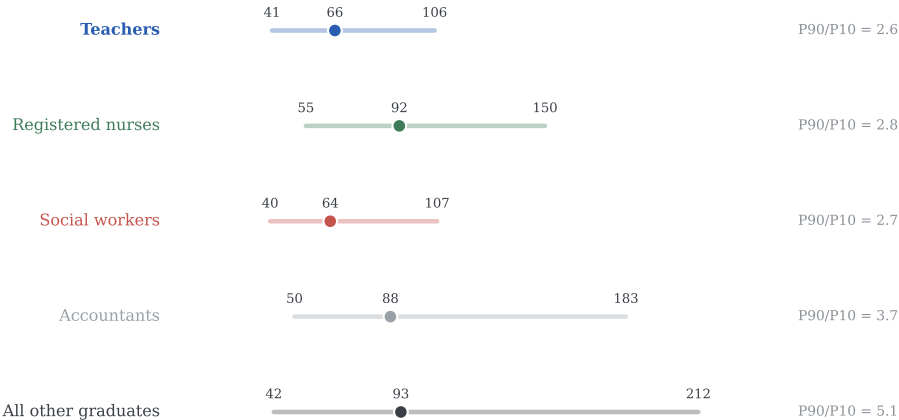
Teachers vs. the college labor market



Teachers vs. the comparable professions

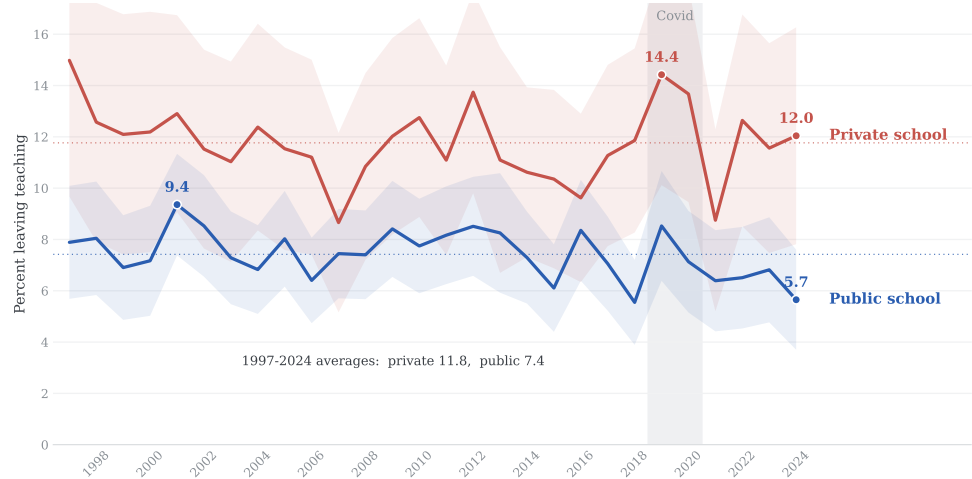


Message 2 · A compressed pay scale

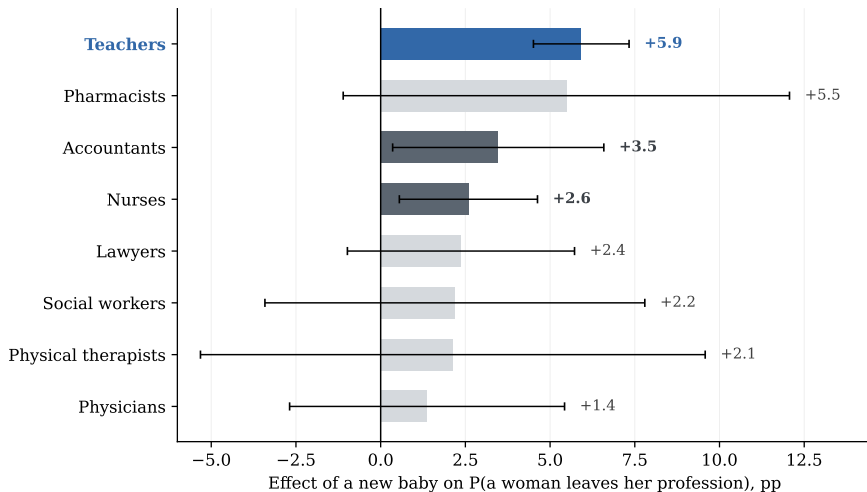


P10–median–P90 of annual earnings, \$1,000 of 2024.

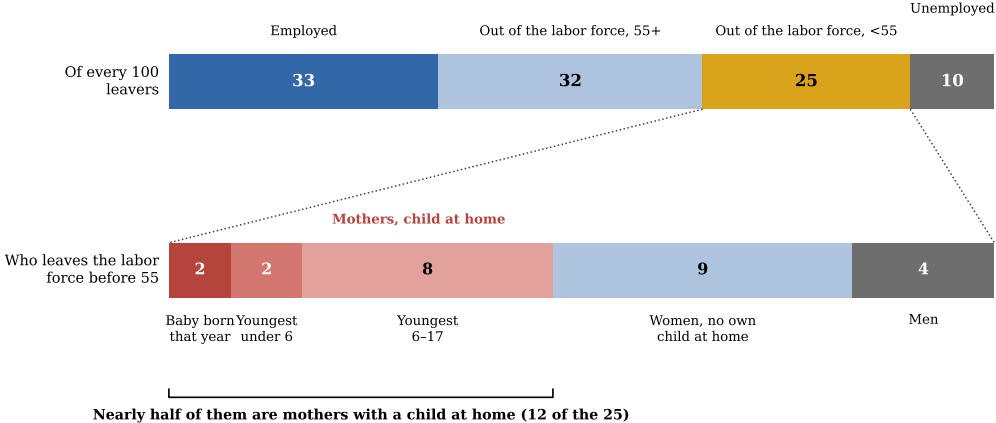
Message 2 · Private schools lose teachers at twice the public rate



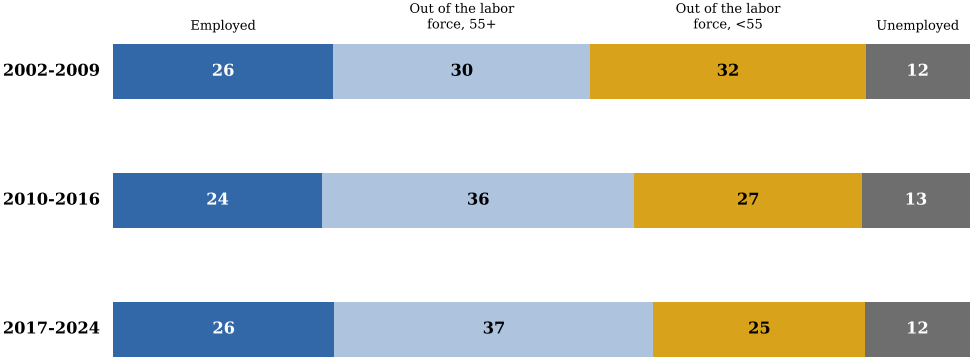
Message 3 · Childbirth moves teachers out more than other professionals



Message 3 · Who leaves the labor force before 55

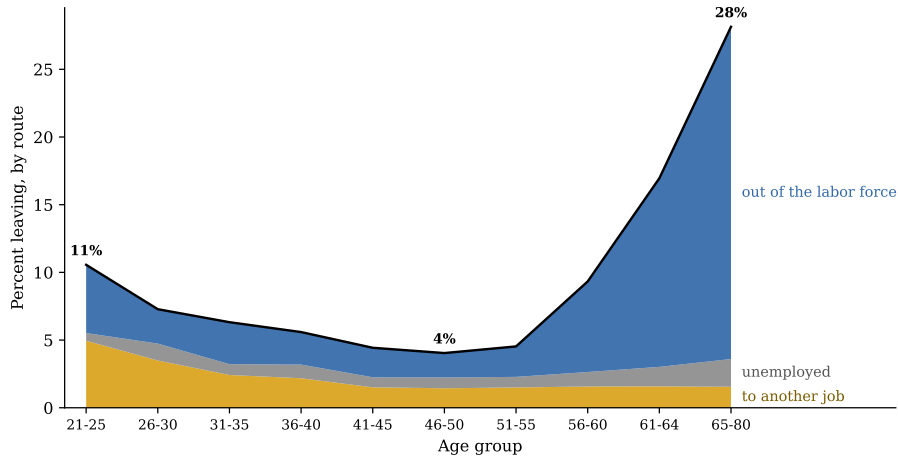


Message 4 · Most leavers exit employment, not to another job

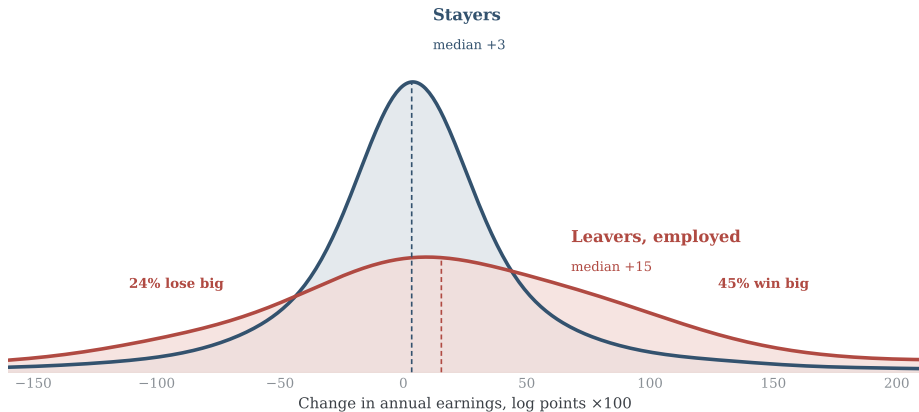


Destinations of leavers, percent of each period's exits.

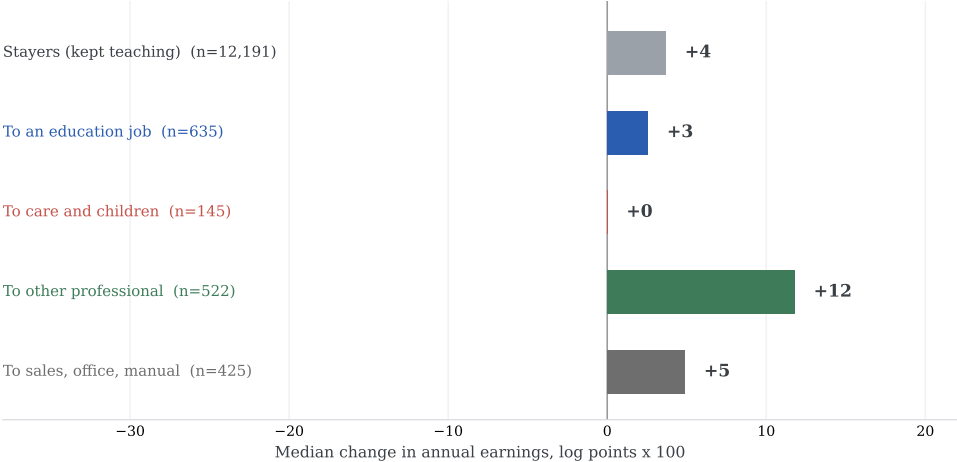
Message 4 · Routes out, across the career



Message 4 · Leavers who stay employed gamble on earnings



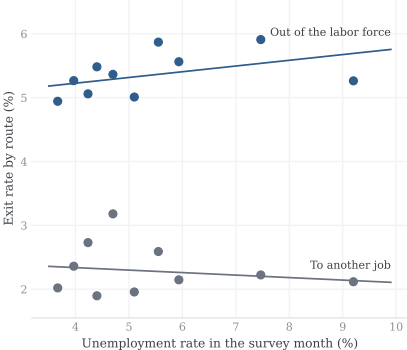
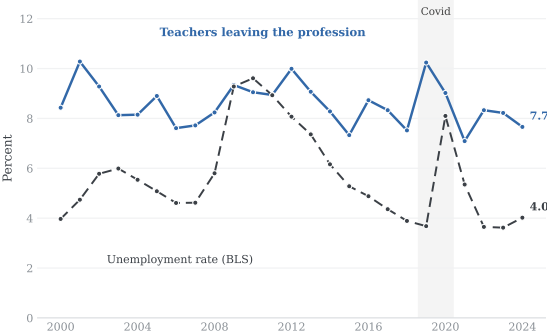
Message 4 · Earnings changes, by destination



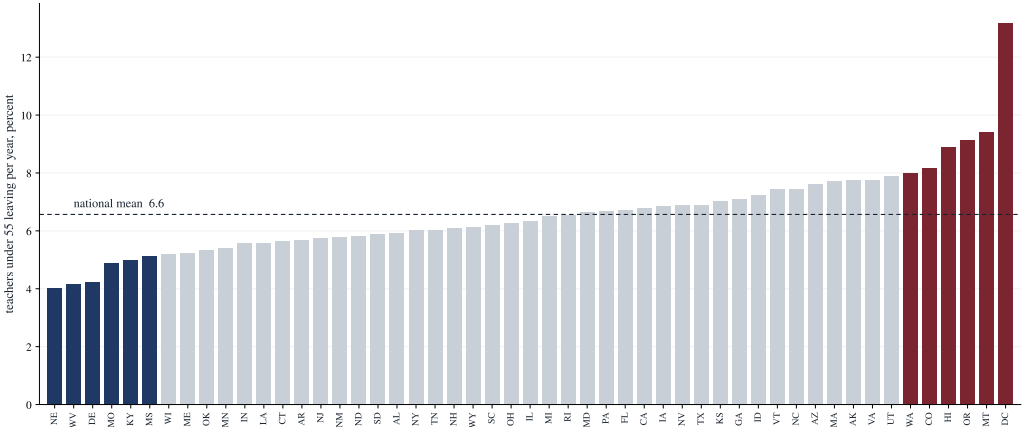
Message 4 · Every destination occupation we observe

EDUCATION, NON-TEACHING — 23%		PROFESSIONAL & MANAGERIAL — 32%		SALES, OFFICE & MANUAL — 36%	
8.1 Postsecondary teachers	9.7 Managers, all other	0.4 Chief executives	0.0 Miscellaneous social scientists...	0.6 Legal support workers, all other	0.3 Billing and posting clerks
4.3 Tutors	1.9 Coaches and scouts	0.4 Dietitians and nutritionists		0.6 Other healthcare support workers	0.3 Occupational health and safety s...
4.2 Teaching assistants	1.7 Architects	0.4 Information security analysts		0.6 Laborers and freight, stock, and...	0.3 Packaging and filling machine op...
3.1 Education administrators	1.5 General and operations managers	0.3 Claims adjusters, appraisers, ex...		0.6 Animal caretakers	0.2 Laundry and dry-cleaning workers
1.5 Librarians	1.3 Business operations specialists...	0.3 Physical therapists	2.8 Secretaries and admin. assistants	0.5 Insurance claims and policy proc...	0.2 Physical therapist assistants an...
1.3 Other instruction and library wo...	1.2 Other mathematical science occup...	0.3 Food service managers	1.8 Retail salespersons	0.5 Personal care aides	0.2 Janitors and building cleaners
0.2 Archivists, curators, and museum...	1.1 Writers and authors	0.3 School psychologists	1.8 Landscaping and groundskeeping w...	0.5 Bill and account collectors	0.2 Payroll and timekeeping clerks
	0.9 Broadcast, sound, and lighting t...	0.3 Editors	1.6 Exercise trainers	0.4 Waiters and waitresses	0.2 Nursing assistants
	0.8 Chemical technicians	0.3 Transportation, storage, and dis...	1.4 Supervisors of personal care and...	0.4 Data entry keyers	0.2 Helpers—production workers
	0.7 Social and community service man...	0.2 Meeting, convention, and event p...	1.3 Taxi drivers	0.4 Hairdressers, hairstylists, and...	0.1 Hosts and hostesses, restaurant...
	0.6 Property, real estate, and commu...	0.2 Training and development managers	1.3 Police officers	0.4 Real estate brokers and sales ag...	0.1 Sales representatives, wholesale...
1.8 Educational and career counselors	0.6 Photographers	0.2 Accountants and auditors	1.2 Executive secretaries	0.4 Customer service representatives	0.1 Security guards and gambling sur...
1.2 Social and human service assista...	0.6 Actors	0.2 Other healthcare practitioners a...	1.1 Stockers and order fillers	0.4 First-Line supervisors of office...	0.1 Chefs and head cooks
1.1 Mental health and substance abus...	0.6 Therapists, all other	0.2 Graphic designers	1.0 Other life, physical, and social...	0.4 Manicurists and pedicurists	0.1 Receptionists and information cl...
1.1 Social workers, all other	0.5 Lodging managers	0.2 Training and development special...	1.0 First-Line supervisors of non-re...	0.4 Other entertainment attendants a...	0.1 Computer, automated teller, and...
0.8 Childcare workers	0.5 Construction managers	0.2 Materials engineers	0.9 Title examiners, abstractors, an...	0.4 Construction laborers	0.1 Maintenance and repair workers...
0.6 Mental health counselors	0.5 Miscellaneous health technologi...	0.2 Civil engineers	0.8 Personal care and service worker...	0.4 Office and administrative suppor...	0.1 Office clerks, general
0.5 Counselors, all other	0.4 Other psychologists	0.1 Other designers	0.7 Fast food and counter workers	0.3 Cooks	0.0 Stationary engineers and boiler...
0.5 Substance abuse and behavioral d...	0.4 Other engineering technologists...	0.1 Wholesale and retail buyers, exc...	0.7 First-Line supervisors of retail...	0.3 Bookkeeping, accounting, and aud...	0.0 Tellers
0.4 Other community and social servi...	0.4 Management analysts	0.1 Physical scientists, all other	0.7 Couriers and messengers	0.3 Miscellaneous agricultural worke...	0.0 Ushers, lobby attendants, and ti...
0.4 Probation officers and correctio...	0.4 Human resources workers	0.1 Project management specialists	0.7 Statistical assistants	0.3 Travel agents	
	0.4 Veterinarians	0.1 Dietetic technicians and ophthal...	0.7 Paralegals and legal assistants	0.3 Sewing machine operators	
		0.0 Floral designers	0.6 Cashiers	0.3 Bartenders	
		0.0 Farmers, ranchers, and other agr...	0.6 Recreation workers	0.3 Conveyor, dredge, and hoist and...	
		0.0 Landscape architects	0.6 First-line supervisors of landsc...	0.3 Boilermakers	

Message 4 · And exits out of employment move with the cycle

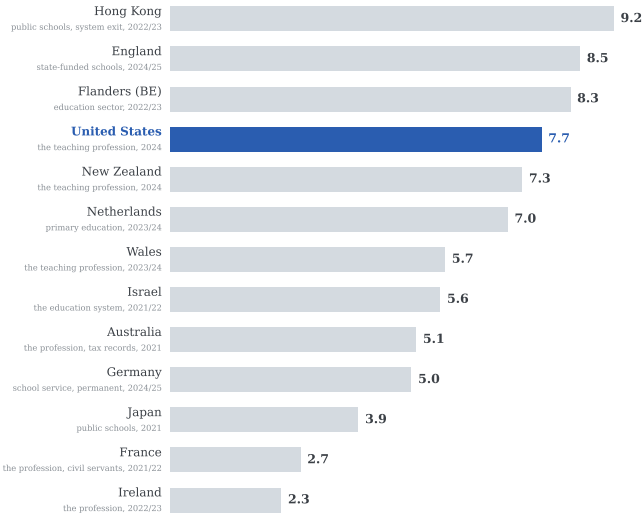


Message 5 · Attrition differs widely across states



State means, 1997–2024.

The US rate, next to the systems that publish one



Conclusion

- A labour force survey can measure teacher attrition at the person level, yearly, over three decades, and it replicates the official picture of the workforce.
- The definition of a leaver is a first-order choice: the same records yield rates from 5 to 19 percent per year.
- What the data say: attachment, pay and pensions, and life-cycle events correlate with leaving; demographics much less.
- The design travels: the same construction is feasible in the labour force surveys of Italy, Germany, the UK, France, Spain, Brazil and Mexico.